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Unit – 3 : Session – 12 : SLO – 2 | DEQUE — Insertion Operations

■ I. Insert at Right of Double-Ended Queue

```
• void insert_right() {  
• int val;  
• printf("\n Enter the value to be added:");  
• scanf("%d", &val);  
• if ((left == 0 && right == MAX-1) || (left == right + 1)) {  
• printf("\n OVERFLOW");  
• return;  
• }  
• if (left == -1) {  
• left = 0;  
• right = 0;  
• }  
• else {  
• if (right == MAX-1)  
• right = 0;  
• else  
• right = right + 1;  
• }  
• deque[right] = val;  
• }
```

■ II. Insert at Left of Double-Ended Queue

```
• void insert_left() {  
• int val;  
• printf("\n Enter the value to be added:");  
• scanf("%d", &val);  
• if ((left == 0 && right == MAX-1) || (left == right + 1)) {  
• printf("\n Overflow");  
• return;  
• }  
• if (left == -1) {  
• left = 0;  
• right = 0;  
• }  
• else {  
• if (left == 0)
```

- left = MAX-1;
- else
- left = left - 1;
- }
- deque[left] = val;
- }